Memory & Cognition
https://doi.org/10.3758/s13421-025-01807-9

Differential contributions of global and local object landmarks in human wayfinding behavior

Wen-Jing Lin[1] · **Erik Chihhung Chang**[2]

Received: 17 June 2024 / Accepted: 21 September 2025
© The Psychonomic Society, Inc. 2025

Abstract
Wayfinding success hinges on the effective processing of landmarks and environmental structures, yet studies examining the functional roles of different landmark types have yielded inconsistent results. This study explored the joint effects of landmark types and environmental structure regularity on wayfinding efficiency by factorially manipulating these variables. Participants navigated through environments with either regular or irregular structures, containing either global or local landmarks. We measured travel durations and paths, finding that both varied according to landmark type and environmental regularity. Notably, during the learning trials when landmarks were visible, both types improved wayfinding in irregular environments but not in regular ones, indicating that the effectiveness of salient cues depends on environmental uncertainty. Conversely, in the probe trial where landmarks were invisible, wayfinding in all environments improved following exposure to local landmarks, but only irregular environments benefited from global landmarks. These findings imply that local and global landmarks are processed and utilized differently, supporting a hierarchical processing model that weights different types of information based on their availability and reliability.

Keywords Wayfinding · Global landmarks · Local landmarks · Spatial navigation

Introduction

Wayfinding is the ability to determine and maintain a cruising path from one's current location to an initially invisible goal location, which involves knowing where to go, where I am, and how to get there (Maguire et al., 1998). It is a complex cognitive ability that consists of several cognitive components (Wolbers & Hegarty, 2010), including at least the spatial memory of targets and landmarks, representation of self positions, egocentric and allocentric spatial representations, working memory, attention, and motor abilities (Allen, 1999). Although sensory inputs from tactile, vestibular, olfactory, kinematic, proprioceptive, and visual modalities can all be helpful for the wayfinding behavior, humans rely most heavily on visual perception (Allen, 1999). Moreover, this process can be understood within a hierarchical processing framework, in which spatial information is organized at multiple, nested levels. At the local level, individuals encode fine-grained, context-specific cues that support immediate navigational decisions, while higher-level, global representations provide a broader, survey-like understanding of the overall environment (Chrastil & Warren, 2014; Weisberg & Newcombe, 2018).

When using visual perception for wayfinding, landmarks provide an essential source of spatial references and enhance wayfinding greatly (Jacobs et al., 1997; Jansen-Osmann, 2002; Jansen-Osmann & Fuchs, 2006; Kelly et al., 2008; Maguire et al., 1998; Sandstrom et al., 1998). Landmarks are salient objects or geometric features in the environment that people can easily observe. According to their visual properties and the manner in which they are processed by the individuals, landmarks can be further divided into two major types—namely, geometric landmarks and object landmarks. Geometric landmarks are salient features of the environmental structures or surfaces (Gallistel, 1990), such as T-junction intersections, crossroads, or the shape of circumference walls. In contrast, the object landmarks are the salient objects within the environment that can be easily distinguished from the background, such as the Eiffel Tower,

✉ Erik Chihhung Chang
audachang@g.ncu.edu.tw

[1] Independent Researcher, Taipei City, Taiwan

[2] National Central University, Room 601, Science Building 5, 300 Jhongda Rd., Jhongli District, Taoyuan City 32001, Taiwan

Published online: 27 October 2025

a statue, or a streetlamp. The object landmarks can be further divided into global and local object landmarks. Global landmarks, such as skyscrapers, are visible from almost any vantage point in the environment and are instrumental in establishing overall orientation and constructing a large-scale cognitive map (Steck & Mallot, 2000). In contrast, local landmarks—like a streetlamp or a corner pub—are typically smaller and visible only from specific locations, providing detailed cues for immediate navigational adjustments.

We can consider examples in daily life drawn from both natural settings and man-made environments. Global landmarks in natural settings typically include prominent, widely visible features such as mountain peaks, rivers, or expansive areas of distinctive vegetation. For example, a distant mountain peak or a river running through a forest can serve as stable orientation points visible from multiple vantage points, helping navigators maintain a sense of general direction over larger distances. In contrast, local landmarks might include smaller-scale, location-specific natural elements such as a uniquely shaped tree, a distinctive rock formation, or a particular clearing in dense woods. These local features provide detailed spatial references visible only within restricted areas and are particularly useful for immediate route decisions or adjustments. Together, such everyday examples illustrate the complementary roles of local and global cues, a relationship formalized in the hierarchical processing model.

The hierarchical processing model proposes that spatial information is organized into nested levels, where local landmarks provide fine-grained anchors for route-based decisions, and global landmarks support coarse-grained survey knowledge that integrates routes into broader cognitive maps (Chrastil, 2012; Chrastil & Warren, 2014; Weisberg & Newcombe, 2018). Early theories already distinguished survey-based from route-based knowledge (Thorndyke & Hayes-Roth, 1982), and subsequent work has shown that individuals often begin with reliance on proximal local cues before gradually incorporating distal global cues to achieve large-scale orientation (Peer & Epstein, 2025; Strickrodt et al., 2019). Importantly, hierarchical organization is not limited to landmarks alone—models emphasize the integration of geometric structure, path integration, and multimodal cues (Burgess, 2008; Wang & Spelke, 2002; Wiener & Mallot, 2003). This dual-level structure helps explain how navigators flexibly shift between immediate route guidance and higher-order spatial planning, why both local and global cues are essential for robust wayfinding, and how navigation relies on dynamic interactions between nested spatial representations within distributed brain networks (Clark et al., 2018; Epstein et al., 2017; Wolbers & Büchel, 2005).

Besides object landmarks, geometric cues, such as the spatial layout of boundaries, walls, and environmental structures, play a foundational role in spatial navigation and wayfinding. These cues are often processed allocentrically, meaning they provide a global framework for orientation that is independent of the navigator's position (Cheng, 1986; Gallistel, 1990). Research has shown that humans and animals rely on geometric properties of an environment, such as distances and angles between boundaries, to establish a sense of direction and location (Cheng & Newcombe, 2005). For example, the geometric module hypothesis suggests that geometric information is encoded automatically and can be used to reorient oneself after disorientation (Hermer & Spelke, 1994). Such cues are especially critical in environments where object landmarks are sparse or ambiguous because they provide stable and invariant reference points. The integration of geometric cues with object landmarks is also supported by theories of spatial cognition that emphasize the hierarchical nature of spatial representations. According to these theories, geometric properties form the backbone of cognitive maps, while object landmarks serve as supplementary features that enhance navigation efficiency (Wang & Spelke, 2002; Wolbers & Wiener, 2014). This hierarchical integration allows navigators to flexibly adapt their strategies depending on the availability and reliability of different types of cues.

Building on this perspective, multimodal navigation theories further suggest that geometric and landmark cues are dynamically integrated with other sensory modalities, such as vestibular, proprioceptive, and auditory information, to support flexible wayfinding strategies (Wolbers & Hegarty, 2010). Within the hierarchical model, this integration can be understood as a layered process: local object landmarks and egocentric path integration guide immediate route choices, while global landmarks and allocentric geometric frameworks anchor broader spatial orientation. When environments are predictable and structurally regular, geometric cues may dominate as stable, higher-order references; in contrast, in irregular or uncertain contexts, local object landmarks and body-based cues often become more critical for anchoring navigation decisions. Thus, the relative contribution of different cue types is not fixed but context-dependent, reflecting a dynamic weighting process within the hierarchical and multimodal navigation system.

Successful navigation involves the flexible and dynamic integration of multiple sources of information, including geometric cues, object landmarks, and path-based (egocentric) strategies. This perspective aligns with dual-process models of spatial navigation, which suggest that humans use both allocentric (map-like) and egocentric (route-based) representations depending on task demands and environmental constraints (Chrastil & Warren, 2014). For instance, when navigating an unfamiliar environment with irregular structures, individuals may rely more heavily on global geometric cues to establish orientation. Conversely, in environments with prominent object landmarks, egocentric strategies may dominate.

In a natural environment, global and local objects landmarks almost always coexist and may conjointly guide wayfinding. This makes it challenging to discern how global and local landmarks contribute to our wayfinding behavior, respectively. There are only a handful of studies that systematically compared the benefits brought by global and local object landmarks in wayfinding (see Yesiltepe et al., 2021, for a comprehensive review), and the results were inconsistent in terms of whether global object landmarks could benefit wayfinding performance. Some studies found that global object landmarks provide no benefit for wayfinding (Meilinger et al., 2015; Ruddle et al., 2010; Yesiltepe et al., 2020) while others found that they do (Credé et al., 2019; Gardony & Taylor, 2011; Lin et al., 2012; Steck & Mallot, 2000).

Steck and Mallot (2000) was the first study to compare the role of global and local landmarks on human wayfinding behavior. In their study, participants acquired the knowledge of spatial layout by freely navigating in a virtual environment with both global and local landmarks. A testing phase then followed where the locations of local or global landmarks were selectively altered to single out the impact associated with altering location of each type of landmarks. Steck and Mallot (2000) found highly idiosyncratic reliance on landmarks: Individual's wayfinding performance, as indicated by prolonged travel duration and distance, was compromised either by the displacement of learned global landmarks, local landmarks, or both. In addition, participants who had a preferred type of landmark performed equally well when only the nonpreferred type of landmark was available. It seems that both representations of types of landmarks were stored in the wayfinding system and can be used reasonably well depending on the context, though one may lean toward using one of them.

While these findings provided strong evidence that both global and local landmarks are encoded and can guide navigation, they also highlighted an important ambiguity in how such cues are integrated. The idiosyncratic reliance observed by Steck and Mallot (2000) could reflect genuine individual differences in strategy, but it could also stem from the specific way their paradigm presented landmarks simultaneously. With both cue types available throughout learning, participants may have formed blended spatial representations in which global and local landmarks were not cleanly separable. This makes it unclear whether the observed flexibility was due to a redundant encoding of distinct landmark systems, or to a single, interdependent representation in which the relative contribution of each cue type cannot be disentangled.

To clarify these ambiguities, the present study separated the availability of global and local landmarks during training, allowing us to assess their distinct contributions to spatial learning. We further manipulated the structural regularity of the environment, reasoning that the predictability of spatial layout may interact with landmark type to shape wayfinding performance. Preliminary observations from our laboratory suggested that local landmarks are especially effective in irregular layouts, leading us to hypothesize that environmental structure systematically modulates the relative utility of global and local cues.

In terms of evaluating performance, travel duration is a critical metric in spatial navigation research, as it reflects both wayfinding efficiency and the cognitive processes underlying successful navigation (Caduff & Timpf, 2007; Klippel & Winter, 2005). Distinguishing between wayfinding (the cognitive and decision-making aspects) and travel (the actual locomotion through space) highlights the dual role of travel duration in capturing these interconnected elements (Allen, 1999). Specifically, travel duration provides insight into how participants develop mental representations of space and implement navigational decisions. Across sequential trials, changes in travel duration reflect the refinement of cognitive maps and spatial memory as participants become more familiar with an environment. This measure aligns with theoretical frameworks suggesting that spatial navigation shares neurocognitive mechanisms with episodic memory and mental simulation, including self-projection and scene construction (Chrastil & Warren, 2014; Weisberg & Newcombe, 2018). By tracking improvements in travel duration across trials, researchers can observe how these cognitive processes evolve in response to environmental cues.

Overview and hypothesis

To examine whether global and local object landmarks contribute differently to the wayfinding process and the mechanisms underlying the difference, we constructed virtual environments for participants to learn the locations of targets when only global landmarks, local landmarks, or no object landmark at all are available. Participants navigating with local object landmarks *should* demonstrate shorter travel durations and distances, compared to those navigating with global landmarks or without any object landmarks. Furthermore, we manipulated the structural regularity of the environment which has been found to impact navigation efficiency across age groups (Merhav & Fisher-Gewirtzman, 2023), and predict that the difference in wayfinding efficiency between local and global landmarks will vary across different structural regularity. Specifically, in environments with irregular structures, participants will rely more on geometric cues when only global landmarks are available, suggesting a flexible approach of using available cues in wayfinding. These hypotheses aim to extend the findings of Steck and Mallot (2000) by isolating the effects of global and local landmarks on wayfinding performance and

examining how these effects may be modulated by the structural characteristics of the environment.

A secondary goal of this experiment was to examine whether the visibility of the target influences wayfinding efficiency. This is achieved by making the target invisible at the last (probe) trial. If the participant really formed memory of the exact target location, his or her travelling duration should not differ much from previous trials when the target remained visible. With the last probe trial, one has the opportunity to assess the strength of pure spatial representation under different combinations of landmark types and structural regularities.

Method

Participants

Twenty-six right-handed participants with normal or corrected-to-normal vision were recruited from the campus of National Central University (NCU), Taoyuan City, Taiwan. There were 13 men (mean age = 21.4 years, *SD* = 3.3) and 13 women (mean age = 19.9 years, *SD* = 3.8). The sample size was determined based on outcomes of a previous study (Lin et al., 2012), which compared gender differences in wayfinding performance in virtual environments with local and global landmarks and recruited 30 participants. Based on its effect size for the main effect of landmark types, we determined that the current sample size was sufficient to reach significance for landmark types. Those who reporting a history of virtual sickness were excluded from participation. Informed consent was obtained from all participants prior to the experiment, in accordance with a protocol approved by the Research Ethics Committee of National Taiwan University, Taipei, Taiwan.

Apparatus and materials

The 3D virtual environment was constructed with 3ds Max 9 (Autodesk®). The control of navigation within the virtual environment was accomplished with custom software written in C++ language and the WorldToolkit toolbox on an IBM compatible PC. The virtual environment was presented on a 22-in. CRT monitor in a nonimmersive fashion. Participants viewed the CRT from a distance of 57 cm, approximately. They perceived the virtual environment in the first person viewpoint, and they could move forward or backward, and turn rightward and leftward by pressing four different buttons, respectively, on a computer keyboard.

From a first-person perspective on the 22-in. CRT monitor, one spatial unit in the virtual environment was estimated to be 4.2 cm in height, corresponding to 4.2° of visual angle at the viewing distance in the current study. The virtual environment was a square consisting of 750 × 750 spatial units and was surrounded by boundary walls that were four units high. Within the environment there were several interconnected roads (five units wide). Neither the texture of the wall nor the ground contained distinguishable features to serve as landmarks. The background scene outside the maze was uniformly colored in flat blue. Every time a predesignated button was pressed, the field of view (FOV) changed one unit in heading orientation.

Inside each virtual environment, a cartoon picture of a common object (airplane, sailboat, and car for all of the three conditions, respectively) served as the target which the participants could not see from the start position. A small icon showed the target picture on the bottom of the screen to remind the participants what they were looking for. The icon remained visible until the participant approached within three units away from the target (see Fig. 1, top row).

There were three landmark conditions:

(1) The global-landmark condition,
(2) The local-landmark condition, and
(3) The no-landmark control condition (Fig. 1, bottom row).

The global-landmark condition presented 12 physically large architectures (beacon, cabin, electric tower, gazebo, skyscraper, arch-bridge, crane, mill, temple, tent, mansion, and billboard) that surrounded the boundary walls and were so large that they were visible all over the environment (as can be seen in Fig. 1, top right and middle right). The local landmark condition presented 12 cartoon pictures of common objects (car, butterfly, cake, cabinet, piano, tree, telephone, turtle, umbrella, vase, watch, and zebra) that were on predetermined positions on the walls or structures and were visible only from limited locations. At least one local landmark can be seen at every position and orientation inside the virtual environment. The no-landmark condition did not have any object landmarks.

We deliberately made the physical scale of the landmarks aligned with their semantic associations: global landmarks were rendered as very large structures, consistent with their implied size, while local landmarks were depicted as smaller, everyday objects. This design choice ensured that the distinction between global and local landmarks reflected not only differences in spatial scope but also their ecological plausibility as navigational cues.

The landmark selection process was explicitly designed to differentiate landmark types through visibility characteristics and structural placement. Global landmarks were positioned beyond navigable boundaries with omnidirectional visibility, consistent with operational definitions of global cues as orientation aids visible from multiple vantage points (Newman et al., 2007; Steck & Mallot, 2000). Local landmarks

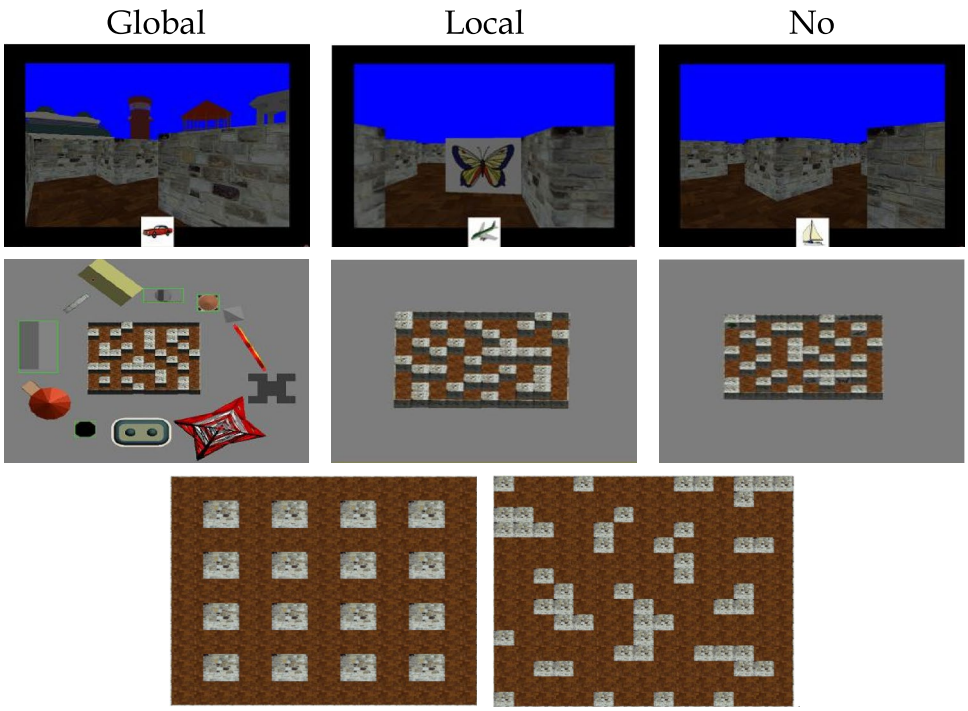

Fig. 1 First-person and top-down views of virtual mazes. The upper row demonstrates the first-person views of mazes with global, local, and no landmarks, respectively, as the participant was navigating the maze. The small icons at the middle-bottom of each graph indicate the target of a trial. The middle row demonstrate the top-down view of the mazes, where the relative spatial arrangement of the landmarks are illustrated. The bottom row contrasts environment with regular (left) and irregular (right) structures

(12 common objects like telephone and zebra) were embedded within the traversable environment at junction-specific locations, visible only within limited spatial contexts—a design choice aligning with route-level cue definitions in virtual navigation research (Geiger et al., 1997; Weisberg & Newcombe, 2018). This visibility-based categorization intentionally avoided confounds from color or size variations by maintaining consistent scaling: global landmarks averaged 15 times larger than local counterparts to ensure continuous visibility without introducing abrupt saliency gradients. While perceptual saliency was not systematically controlled, the structural differentiation between landmark types (external vs. embedded placement, persistent vs. localized visibility) created distinct functional categories that mirror real-world urban wayfinding contexts where global cues (e.g., towers) and local cues (e.g., storefronts) naturally differ in spatial relationships rather than visual prominence (Caduff & Timpf, 2007). The no-landmark condition's complete absence of object landmarks provided a baseline for isolating structural versus landmark-dependent navigation strategies.

To avoid the potential confounding that practice effect associated with a particular structural layout of the environment would facilitate learning, three environments with distinct layout but equivalent structural complexity had been created. This results in nine different combinations of virtual environments and landmark conditions, which were counterbalanced across participants. The length of the shortest path (200 spatial units) from the start position to the target in each testing trial was equal in all conditions.

Two levels of environmental regularity were implemented in the virtual environments. The regular environment featured uniformly arrayed blocks in a predictable checkerboard pattern (see Fig. 2A), providing a clear, systematic layout. In contrast, the irregular environment was intentionally designed to deviate from this order by arranging blocks and pathways in a random, asymmetrical manner (see Fig. 2B). Although no strict quantitative metric was applied to define "irregularity," these layouts were crafted to reflect the unpredictable, non-uniform characteristics observed in real-world settings—such as historic town centers or organically developed urban areas. In both environment types, four targets were positioned, and the total traversable area was carefully matched, ensuring that any differences in navigational performance stem from the structural regularity rather than variations in spatial extent.

Procedure

Participants were instructed find targets in both the regular and irregular environments, and each environment contained three landmark conditions. They navigated through two environments in separate sessions, with at least a 3-min

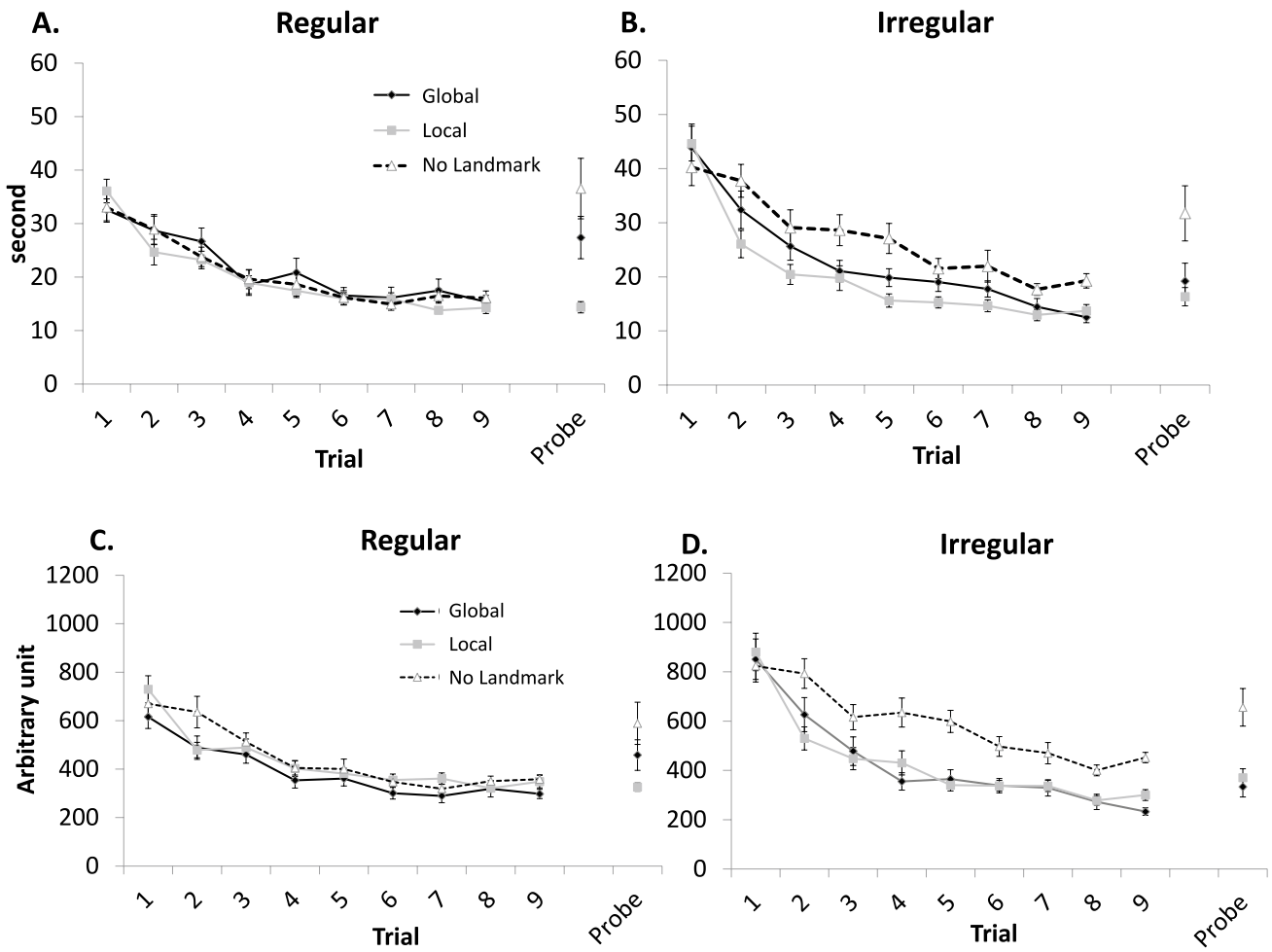

Fig. 2 Travel duration (in second; **A** & **B**) and travel path (in arbitrary unit; **C** & **D**) as a function of the Structural Regularity, Landmark Types, and Trial Order. The error bars indicate standard errors

break between sessions for the participant to rest and the experimenter to set up the second session.

The regular environment was always presented before the irregular environment. We chose a fixed order for three methodological reasons. First, asymmetric carryover: Beginning with an irregular layout induces strong, idiosyncratic search strategies (e.g., exhaustive corridor scanning, "safe-turn" heuristics) that tend to persist and contaminate subsequent behavior even when the environment becomes easier. In contrast, beginning with a regular grid establishes a neutral baseline of basic task control (controls, timing, response mapping) without forcing compensatory strategies that would later mask landmark effects. Second, measurement sensitivity: Irregular-first markedly increases the risk of floor performance (long paths/times, aborted trials), compressing between-condition differences and reducing power to detect landmark effects; regular-first keeps early trials within a measurable dynamic range so that differences among landmark types can be observed from the outset. Third, participant viability: Irregular-first elevates disorientation and frustration, which amplifies variance unrelated to the constructs of interest and can result in early withdrawal or conservative responding. Together, these considerations make a fixed regular-first order the most conservative choice for isolating the effect of landmark information on wayfinding efficiency.

The order of landmark conditions within each session was counterbalanced across participants. Participants were not explicitly informed of separate learning and testing phases. Instead, they completed a sequence of wayfinding trials in each landmark condition. For each condition, participants navigated through a virtual environment in search of a designated target object. Across the initial nine trials, landmarks remained visible, allowing participants to gradually learn the spatial relationships between the environment, landmarks, and the target location. These trials functioned as implicit learning trials, during which participants refined

their navigation strategies and improved their efficiency in reaching the target.

Each trial had a 150-s time limit, ensuring participants had sufficient opportunity to explore the environment while preventing excessive search time. The target remained in a fixed location across all nine trials within a given condition, reinforcing spatial learning over repeated exposures. Following the nine trials, a probe trial was conducted. In this final trial, one of the four possible targets was randomly selected, but unlike previous trials, the target was invisible. This condition was designed to assess whether participants had formed a stable cognitive representation of the environment, allowing them to navigate successfully using only their memory of the target's location relative to the landmarks and environmental layout. Throughout both the learning trials and the probe trial, all target locations remained unchanged, ensuring that any observed improvements in navigation reflected true spatial learning rather than dynamic target-seeking behavior. The probe trial provided a critical measure of how well participants retained and applied their spatial knowledge when deprived of direct visual cues.

Data analysis

Both *travel duration* and *trave path* data are subject to the same statistical analysis. For the *learning trials* (i.e., Trials 1–9 where targets were visible), the travel duration data from the wayfinding trials were subject to a three-way repeated-measures analysis of variance (ANOVA), with landmark types (global/local/no landmarks), structural regularity (regular/irregular), and trial order (1–9) as within-subject variables. For the probe trials when the target was not visible, the outcomes were subject to a two-way repeated-measures ANOVA, with landmark types and structural regularity as within-subject variables. Partial eta-squared was reported to indicate the effect sizes of the ANOVAs.

Results

Learning trials: Travel duration

For performance in the visible trials (Trials 1~9), we examined the influence of structural regularity, landmark types, and trial order on travel duration using a three-way repeated-measures ANOVA. The outcome revealed significant main effects for structural regularity, $F(1, 25) = 15.16$, $MSE = .624$, $p = .001$, $\eta_p^2 = .38$; landmark types, $F(2, 50) = 7.46$, $p = .001$, $MSE = .166$, $\eta_p^2 = .23$; and trial order, $F(8, 200) = 122.75$, $p < .001$, $MSE = 1.946$, $\eta_p^2 = .83$. Specifically, these main effects indicate that participants took longer to traverse irregular environments ($M = 23.4$ s) compared with regular ones ($M = 20.8$ s). Travel duration was also shorter in the local condition ($M = 20.2$ s) relative to the control ($M = 23.9$ s) and global ($M = 22.2$ s) conditions. Additionally, the average travel duration across all conditions decreased from 38.4 s in the first trial to 15.2 s in the ninth trial.

Significant two-way interactions were observed between structural regularity and landmark types, $F(2, 50) = 4.13$, $p < .05$, $\eta_p^2 = .14$. In irregular environments, participants took longer in the no-landmark condition ($M = 27.2$ s) compared with both the global ($M = 26.2$ s) and local ($M = 23.6$ s) conditions (p values $< .05$). In contrast, in regular environments travel durations were comparable across conditions (no landmark: $M = 22.2$ s; global: $M = 21.4$ s; local: $M = 21.2$ s).

The interaction between structural regularity and trial order, $F(8, 200) = 3.8$, $p < .001$, $\eta_p^2 = .13$, was also significant. At the beginning (Trial 1), travel duration was much higher in irregular environments ($M = 42.9$ s) than in regular ones ($M = 33.9$ s). However, with practice across Trials 2–9, this difference diminished, converging to nearly identical levels by the final trials (irregular: 15–32 s; regular: 15–27 s).

The interaction between landmark types and trial order, $F(16, 400) = 1.78$, $p < .05$, $\eta_p^2 = .07$, reached significance. In Trial 1, the three landmark conditions did not differ reliably (local: $M = 40.6$ s; control: $M = 36.6$ s; global: $M = 38.2$ s). Across subsequent trials, however, durations diverged: The no-landmark condition consistently yielded the longest durations, followed by the global condition, with the local condition shortest (no landmark: 17.1–33.3 s; global: 13.9–30.5 s; local: 13.4–25.4 s).

Although the three-way interaction was not significant, $F(16, 400) = 0.81$, $p > .05$, $\eta_p 2 = .03$, we conducted trial-by-trial one-way ANOVAs on landmark types for theoretical interest. In the regular environment, performance did not differ among landmark types at any trial, all $F(2, 50) < 2.14$, p values $> .13$ (Fig. 2A). In the irregular environment, however, significant differences emerged from Trial 2 onward (except Trials 1 and 3), with $F(2, 50) > 3.8$, p values $< .05$ (Fig. 2B). These effects reflected consistently longer travel durations in the no-landmark condition (17.6–37.7 s) compared with the global (12.5–32.4 s) and local (13.0–26.1 s) conditions.

Learning trials: Travel path

The mean length of travel path were depicted in Fig. 2C (regular environment) and Fig. 2D (irregular environment). As demonstrated in the figure, outcomes of the travel path are generally consistent with those of the travel duration. To compare the travel path of the three different types of landmarks for regular and irregular environments, respectively, we conducted one-way ANOVAs on landmark types

at each trial for different structural regularity. In the regular environment, none of the comparison among landmark types reached significance in any trial, all $F(2, 50) < 2.72$, p values >.08, meaning that the length of travel path in no-landmark (287~730 units), global (M = 263~682 units), and local (M = 289~786 units) conditions did not differ from one another (Fig. 2C). However, in the irregular environment, almost all trials, except for Trials 1 and 3, reached significance among landmark types, all $F(2, 50) > 5.25$, p values <.009, η_p^2 >.17, attributable to the longer travel path of the no-landmark (M = 379~854 units) than the global (M = 215~693 units) and local (M = 243~583 units) conditions (Fig. 2D).

Probe trial: Travel duration

The two-way repeated-measures ANOVA on travel duration of the invisible probe trials showed a significant main effect of landmark types, $F(2, 50) = 11.21$, p <.001, η_p^2 =. 31. The no-landmark condition had the longest travel duration (M = 34.1 s), followed by the global condition (M = 23.3 s), and finally the local condition (M = 15.4 s). The main effect of regularity, $F(1, 25) = 2.34$, p >.05, η_p^2 =. 09, and interaction, $F(2, 50) = 1.4$, p >.05, η_p^2 =.05, did not reach significance.

Out of theoretical interests and to further understand these findings, separate one-way ANOVAs were conducted for regular and irregular environments, respectively. In the regular environment, a significant main effect of condition was observed, $F(2, 50) = 8.23$, p <.001, η_p^2 =.25. Post hoc comparisons indicated that travel duration in the local condition (M = 14.3 s) was significantly shorter than in both the global (M = 27.4 s) and control (M = 36.5 s) conditions (both p values <.05). In the irregular environment, the main effect of condition was also significant, $F(2, 50) = 6.63$, p <.01, η_p^2 =.21. Post hoc tests revealed that the no-landmark condition (M = 31.7 s) had significantly longer travel durations compared with both the global (M = 19.2 s) and local (M = 16.3 s) conditions (Fig. 2A–B, Probe).

Probe trial: Travel path

The two-way repeated-measures ANOVA on probe-trial travel path (Structural Regularity × Landmark Type) revealed a significant main effect of landmark type, $F(2, 50) = 11.9$, p <.001, η_p^2 =.322, but no main effect of regularity, $F(1, 25) = 0.01$, p >.05, and no significant interaction, $F(2, 50) = 1.91$, p >.05. Despite the nonsignificant interaction, planned one-way ANOVAs within each environment showed robust effects of condition in both regular and irregular structures. In the irregular environment, post hoc tests indicated that participants traveled significantly shorter paths in both the global (M = 332 units) and local (M = 376 units) landmark conditions relative to the no-landmark condition (M = 650 units), with no difference between global and local cues. In contrast, in the regular environment, local landmarks (M = 329 units) supported significantly shorter travel paths than both global (M = 458 units) and no-landmark (585 units) conditions, while performance in the global and control conditions did not differ. As illustrated in Fig. 2C–D (Probe), these results are consistent with those in travel duration, and demonstrate that local landmarks consistently facilitated the most efficient navigation, but the relative utility of global landmarks emerged selectively in irregular layouts.

Discussion

The objective of the current study was to scrutinize the differential contributions of global and local object landmarks in shaping human wayfinding behavior, and to explore how the structural regularity of the environment modulates the utility of these landmarks. The results reveal distinct patterns of travel duration across different landmark types and structural regularity. In visible trials, travel duration/path was longer for no landmarks compared with local landmarks in irregular environments; no such disparity was observed in regular settings. Conversely, in probe trials when targets were not visible, irregular environments with no landmarks exhibited longer travel durations/paths compared with the other two landmark types, whereas in regular environments, the no landmarks only led to longer travel durations/paths than the local ones.

Furthermore, we also observed that travel duration was longer in the global condition during the initial trials (Trials 2 and 3), but this difference diminished by Trial 4. This temporal pattern suggests that global landmarks may require an extended period of exposure to be effectively utilized for wayfinding. This interpretation aligns with extant literature indicating that global landmarks can indeed facilitate wayfinding, although their primary influence appears to be in enhancing survey knowledge (Credé et al., 2019; Gardony & Taylor, 2011; Löwen et al., 2019). Therefore, while local landmarks may offer immediate navigational benefits, global landmarks could necessitate a longer learning curve but contribute to a more comprehensive spatial understanding.

The significant interaction between structural regularity and trial order suggests that environmental layout influences the rate at which navigators develop efficient strategies. Irregular environments initially impose greater cognitive demands due to their lack of predictable structure, necessitating reliance on local landmarks for route-level decisions (Geiger et al., 1997). However, with repeated exposure, participants appear to integrate global spatial information, enabling comparable performance in both regular and irregular layouts. This progression aligns with hierarchical models of spatial learning, where local landmarks anchor

early navigation efforts before global cues facilitate survey knowledge acquisition (Weisberg & Newcombe, 2018). Similarly, the interaction between landmark types and trial order highlights the complementary roles of local and global landmarks. Local landmarks support immediate decision-making during early trials, while global landmarks become increasingly effective as participants gain familiarity with the environment (Steck & Mallot, 2000). These findings emphasize the dynamic nature of landmark utilization, suggesting that spatial learning involves a gradual shift from reliance on proximal cues to integration of distal, global references.

The present pattern motivates stage-adaptive wayfinding that aligns guidance with users' learning phase and the environment's structure. For first-time users and in irregular layouts, systems should foreground local, decision-point cues—highly distinctive markers placed immediately before and after junctions, consistent turn-by-turn confirmation ("breadcrumbing"), and short-range AR pins or corridor-level wall graphics—because early performance depends on proximal anchors. As familiarity grows, aids should progressively layer global frames that promote survey knowledge: persistent orientation beacons (e.g., district/wing identifiers visible across multiple vistas), stable reference axes (e.g., "north-up" maps or consistent compass cues), and periodic overview inserts that help knit routes into a larger mental map. In regular grid-like settings, the structure itself bears part of the cognitive load, so guidance can be lower-clutter and more minimalist, reserving landmarks for disambiguating visually similar nodes and long-range orientation. Across settings, consistency of iconography, color, and naming should tie local and global elements together; multimodal redundancy (visual + auditory/haptic where appropriate) can assist novices and stressed users; and decision-point placement should be prioritized over mid-segment placement to maximize utility. Collectively, these practices operationalize the hierarchical account we advance: early emphasis on local anchors to reduce immediate choice uncertainty, followed by integration of global cues to support efficient, survey-level navigation with experience.

The current findings extend Steck and Mallot's (2000) framework by demonstrating how environmental structure modulates landmark utilization strategies. Their seminal work showed that individuals differed in whether they relied more heavily on global or local landmarks, suggesting flexible but idiosyncratic strategies. Our results refine this conclusion by revealing that such variability is not only individual but also systematically shaped by structural regularity—a factor absent in their paradigm. In regular, grid-like environments, participants performed equally well with either landmark type, consistent with Steck and Mallot's observation that both cues can support navigation when simultaneously encoded. In irregular layouts, however, local landmarks offered a distinct advantage during visible trials, suggesting that structural unpredictability increases reliance on proximal cues. The probe trials further clarify this relationship: global landmarks became effective only after extended exposure, supporting navigation when targets were no longer visible. This delayed contribution resonates with Steck and Mallot's finding that both landmark types are stored and can be flexibly recruited, but our results add that global cues emerge primarily when environmental unpredictability necessitates broader survey knowledge. Together, these findings recast landmark use as not merely idiosyncratic but as part of a hierarchical, context-sensitive system in which environmental regularity governs the temporal dynamics of local and global landmark utility.

This structural dependency resolves apparent contradictions in prior literature: Geiger et al.'s (1997) local landmark dominance emerges in irregular contexts, while Steck and Mallot's (2000) strategic flexibility applies primarily to regular environments. The testing phase results further clarify this interaction—invisible global landmarks only enhanced navigation in irregular environments, suggesting their role emerges through prolonged environmental exposure rather than immediate perceptual salience. These findings support a hierarchical spatial processing model where local landmarks anchor route-level decisions while global cues contribute to survey knowledge formation, particularly when environmental unpredictability necessitates broader spatial integration(Chrastil & Warren, 2014; Weisberg & Newcombe, 2018).

Findings in the current study challenge the initial hypothesis that consistent differences would be observed between global and local landmarks across both regular and irregular environments. Contrary to expectations, the data revealed a nuanced interplay between the type of landmarks and the structural regularity of the environment. In irregular environments, travel durations were notably longer when navigating with no landmarks compared with local landmarks in visible trials. This observation aligns with the work of Montello and Raubal (2013), who argued that local landmarks often provide more immediate and context-specific cues that facilitate the construction of a mental map, particularly in irregular or complex settings. The increased effectiveness of local landmarks in these contexts may be attributed to their ability to offer "anchor points" that help individuals orient themselves and make decisions at critical junctions (Lynch, 1960).

On the other hand, in regular environments, the absence of significant differences in travel durations between no landmarks and local landmarks suggests that the structural regularity of the environment itself may serve as a navigational aid. This observation is consistent with the "environmental complexity theory" proposed by Weniger et al. (2009), which posits that regular grid-like environments can

reduce cognitive load and reliance on specific landmarks for navigation.

The probe trials, where targets were not visible, further nuanced these findings. In irregular environments, no landmarks were associated with longer travel durations compared with both global and local landmarks. This could be interpreted in light of the "salience model" by Caduff and Timpf (2007), which suggests that landmarks lacking in contextual specificity may not be as effective in guiding navigation, especially when the target is not directly visible. In regular environments, however, no landmarks led to longer travel durations only when compared with local landmarks. This reinforces the notion that local landmarks may serve as more effective cues for navigation, even in structurally regular settings (Jellinger, 1999/2000).

In previous studies that also compared impacts of global and local landmarks on wayfinding, some actually did not implement any control condition (i.e., an environment with no object landmarks available; Credé et al., 2019; Lin et al., 2012), making it impossible to know whether the global landmarks provided any benefits in the wayfinding process as compared with baseline performance. On the other hand, there are also studies lacking structures in their virtual environment. For example, Gardony et al. (2011) build an open virtual arena with no predefined path, and thus without any structures, while participants could travel freely inside it with no obstruction by walls or buildings. Although this design may avoid any interference from structural landmarks, it lacks ecological validity given that predefined streets and paths are inevitable to wayfinding in modern environments. Therefore, findings from the current study provides a sounder contrast between global and local landmarks with better control and ecologically valid virtual environment.

The difficulty participants face in effectively leveraging global object landmarks may be influenced by the structural regularity of the virtual environment. Our current study did not reveal any significant disparities between the utility of global and local landmarks in environments with regular structures. However, this absence of difference invites further exploration. To clarify these ambiguities, future research could incorporate more complex regular environments to isolate the effects of structural regularity on landmark efficacy. In this regard, the work of Jansen-Osmann et al. (2007) provides a valuable reference. Their study utilized a virtual environment with a more intricate regular structure and demonstrated that participants were more adept at using global landmarks for navigation. This suggests that the complexity of the environment may serve as a modulating factor for the effectiveness of global landmarks in wayfinding tasks (Jansen-Osmann et al., 2007).

The current study employed two distinct types of landmarks: global landmarks (large architectures surrounding the boundary walls) and local landmarks (cartoon pictures of common objects on walls or structures). While we did not explicitly manipulate saliency, the inherent characteristics of these landmark types may have influenced navigation efficiency (for a review of impacts of saliency on landmark usage, see Yesiltepe et al., 2021). Global landmarks in our study (e.g., beacon, skyscraper, arch-bridge) were visible from multiple vantage points and could potentially serve as distant orientation cues. Local landmarks (e.g., car, butterfly, cake) were only visible from limited locations and might have functioned as proximal navigation aids. The selection of these stimuli was based on their potential to be easily recognizable and distinguishable within the virtual environment, rather than on a systematic manipulation of saliency. It is important to note that the effectiveness of these landmarks may have been influenced by factors such as their visual distinctiveness, structural placement, and semantic associations (Caduff & Timpf, 2007; Klippel & Winter, 2005; Sorrows & Hirtle, 1999). For instance, a skyscraper (global landmark) might be visually salient due to its size and height, while a cartoon picture of a zebra (local landmark) might be semantically distinctive in an urban environment. Future research could benefit from a more systematic approach to landmark selection and a formal assessment of landmark characteristics to better understand their impact on spatial navigation.

The current results are in line with the concept of a hierarchical representation of spatial relations in large-scale environments (Weisberg & Newcombe, 2018), where detailed representations are formed for local areas as the basis of the cognitive map (e.g., local landmarks), while the connections between these areas are represented more broadly at higher level of the hierarchy (e.g., global landmarks; also see Chrastil & Warren, 2014; Jacobs & Menzel, 2014; Wolbers & Hegarty, 2010). In a regular grid-like structure, higher-level spatial representations contain rough information sufficient for wayfinding. In contrast, an irregular structure requires more detailed information from the lower levels of the hierarchy for effective navigation. Consequently, local landmarks serve as efficient cues in both scenarios, whereas global landmarks are more effective in regular grid-like environments than in irregular ones.

Our findings can also be interpreted within the broader framework of multimodal navigation. The early and stable advantage of local landmarks in irregular environments suggests that when structural predictability is low, navigators rely more heavily on proximal visual cues that can be readily integrated with egocentric body-based information such as vestibular and proprioceptive feedback (Wolbers & Hegarty, 2010). In contrast, global landmarks may require extended exposure before they can be effectively incorporated into an allocentric geometric framework that supports

survey knowledge (Chrastil & Warren, 2014; Weisberg & Newcombe, 2018). This pattern aligns with hierarchical models of spatial cognition, in which cue weighting is context-dependent: local cues and path integration anchor immediate route-level choices, while global landmarks and geometric regularities contribute to higher-level orientation over time (Wang & Spelke, 2002; Wolbers & Wiener, 2014). By situating object landmark effects within this multimodal, hierarchical system, our results highlight how navigators dynamically allocate attention across cue types depending on environmental regularity and uncertainty.

Limitations

In the current study, global landmarks were defined as distinctive architectures, while local landmarks were everyday objects with smaller sizes. However, in addition to the scope of visibility, the global landmarks may have different visual salience from local landmarks. Furthermore, as we did not control color or other perceptual features of different types of landmarks, we cannot rule out the possibility that the difference between global and local conditions was caused by systematical differences of visual salience in the two conditions. Similarly, the global and local landmarks may have incompatible structural and cognitive salience levels. That is, given that the global landmarks were located at a great distance outside the arena and were not accessible while the local ones sit on the walls at the intersections or along the paths, the global and local landmarks here inevitably inherited different structural salience. Finally, the word frequencies of the landmark names were not controlled, nor were responses from participants collected to indicate whether each landmark contains unique meanings. Consequently the cognitive salience of landmarks among different conditions cannot be assessed. Further studies with better control of various kinds of salience can clarify its role in using landmarks during wayfinding in a novel environment.

In addition to the discrepancies in visual salience, another perceptual discrepancy between the landmark conditions may be the scenes presented to the participant during the wayfinding: Only one single landmark can appear on the display in the local condition, while in the global condition, three to five visible global landmarks were always displayed. Though visibility is also one of the definition differentiating global and local landmarks, it is still likely that the higher complexity of the scene and the competition among several landmarks for attention resulted in the inefficiency of using global object landmarks.

We acknowledge that a fixed order raises the possibility that general practice could differentially aid the second (irregular) environment. Three features of our data mitigate this concern:

(i) The local-vs.-global separation in the irregular environment emerged by the second trial and then stabilized, indicating that the critical effect appears before substantial practice can accrue.
(ii) The effect's stability across subsequent trials argues against a gradual practice-driven divergence; instead, it is consistent with an immediate reliance on proximal anchors under structural uncertainty.
(iii) The two environments were topographically distinct, minimizing direct route transfer; any residual order benefit would therefore be strategic rather than mnemonic, which again should grow with practice—contrary to the observed early, flat divergence.

While a counterbalanced or between-subjects order would be ideal, the present design was adopted to avoid asymmetric carryover and floor effects that would have rendered landmark comparisons uninformative in an irregular-first sequence. Future studies can pursue between-subjects counterbalancing with adaptive difficulty (e.g., enlarged corridors, reduced path complexity on initial irregular exposures) to further disentangle order from intrinsic differences in cue use.

A somewhat related concern is the brief, 3-min interval between the regular and irregular environment conditions. This pause was intended to provide a short reset without disrupting task engagement, and similar brief breaks are commonly used in spatial navigation research to reduce immediate interference while maintaining experimental flow. Nevertheless, because we did not formally test whether this interval was sufficient to eliminate carryover effects, this remains a limitation. Future studies could incorporate manipulation checks (e.g., participant ratings or pre–post performance comparisons) and explore longer inter-session intervals to more definitively rule out residual influences between conditions.

The extremely long virtual distance between the observer and the global landmarks can pose another confound: The images of the same landmark observed from various locations could be almost undifferentiable when the viewing orientations of these locations are the same or close. This may keep the participants away from using a single global landmark to locate themselves, and instead they may attempt to "triangularize" themselves by analyzing the spatial relationship among landmarks or between landmarks and environment structures, which makes the task more difficult. This conundrum does not exist in the local condition because the local landmarks could be seen only in a short distance and within a small area, which are more specific and can reduce more uncertainty than the global landmarks. Future studies may adopt more sophisticated technique in creating virtual reality such that global landmarks can be rendered viewpoint-dependent outlooks that brings comparable amount of information as local landmarks.

Fitting and colleagues (2005, 2007, 2009) proposed a category adjustment model to elaborate how the distance between cue and target affect our place learning (cf. category prototype model; Huttenlocher et al., 1991). They dictated that people would have better performance when more than one cue is available or when the distance from target to cues is shorter. In the present study, there were equal number of landmarks in global and local conditions. Therefore, the distance between the target and the nearest landmark may be responsible for the discrepancies between global and local conditions. In the global condition, landmarks were always located outside the arena where participants could travel, while the local landmarks were distributed. The closer proximity between target and landmarks in the local condition than the global condition may have facilitated performance in the local condition. However, this model is agnostic to the influence of geometric structures. In addition, because the interaction between geometric structure and object landmarks had not been studied very well in the literature, we cannot be sure if the prediction of the category-adjustment model would fit the results in the virtual environment in the present study.

While no formal manipulation check was conducted in this study, informal estimation based on the spatial arrangement and pilot observations suggested that each global landmarks were perceptible from more than 90% of the locations in the environment, whereas visibility of each local landmark was restricted to its immediately adjacent neighborhood. This visibility-based distinction aligns with prior research on landmark functions in virtual navigation, where global cues provide omnidirectional orientation and local cues support route-specific decisions, consistent with real-world urban and natural distinctions. In the current study, these classifications were strictly implemented to isolate effects, though we note that real-world landmarks may exhibit more overlap (e.g., a small landmark visible from afar in open terrain). Future work could incorporate explicit checks, such as eye-tracking or posttrial surveys, to validate these designations empirically.

Finally, our current virtual environment did not account for variations in physical effort or energy expenditure, which represent important factors influencing real-world navigational decisions. Specifically, routes that are shorter in distance might sometimes be energetically costly due to challenging terrain, such as hills or rough ground, making longer yet less strenuous paths more advantageous. Future research could address this limitation by incorporating virtual terrain elements (e.g., inclines or obstacles) that systematically vary the physical effort required, enabling investigation into how navigators balance spatial efficiency with energy demands. This would further enhance the ecological validity and generalizability of wayfinding studies.

Conclusions

To conclude, when people learn to orient themselves in a novel environment, local object landmarks facilitate their performance to higher extent than global ones. Nevertheless, global object landmarks can also contribute to the learning of other target locations when one has plenty of exposures to the environment. In sum, by isolating landmark types and incorporating structural regularity, the present study advances Steck and Mallot's (2000) seminal framework from documenting idiosyncratic preferences to demonstrating a hierarchical, context-dependent system of landmark use in human wayfinding. Further studies controlling the salience at various levels and matching the amount uncertainty carried by different types of landmarks shall shed further lights on the cognitive processing underlying wayfinding behaviors.

Author contributions W.-J.L.: conceptualization, methodology, software, data curation, investigation, writing—original draft preparation. E.C.C.: conceptualization, methodology, supervision, funding acquisition, writing—review and editing.

Funding The current study was supported by the National Science and Technology Council, Taiwan, under the grant ID 112-2410-H-008-066 and 108-2410-H008-037-MY4 to Erik Chang.

Availability of data and materials and code availability The datasets generated and/or analyzed during this study, along with the custom programming code utilized, are available from the corresponding author upon reasonable request. This study was not preregistered.

Declarations

Ethics approval and consent to participate Informed consent was obtained from all participants prior to the experiment, in accordance with a protocol approved by the Research Ethics Committee of National Taiwan University, Taipei, Taiwan.

Consent for publication Not applicable.

Financial interests The authors declare they have no financial interests.

Nonfinancial interests None.

Conflicts of interest Both authors declare that they have no conflicts of interest.

References

Allen, G. L. (1999). Spatial abilities, cognitive maps, and wayfinding. In R. G. Golledge (Ed.), *Wayfinding behavior* (pp. 46–80). Johns Hopkins University Press.

Burgess, N. (2008). Spatial cognition and the brain. *Annals of the New York Academy of Sciences.* https://doi.org/10.1196/annals.1440.002

Caduff, D., & Timpf, S. (2007). On the assessment of landmark salience for human navigation. *Cognitive Processing, 9*(4), 249–267. https://doi.org/10.1007/s10339-007-0199-2

Cheng, K. (1986). A purely geometric module in the rat's spatial representation. *Cognition, 23*(2), 149–178. https://doi.org/10.1016/0010-0277(86)90041-7

Cheng, K., & Newcombe, N. S. (2005). Is there a geometric module for spatial orientation? Squaring theory and evidence. *Psychonomic Bulletin & Review, 12*(1), 1–23. https://doi.org/10.3758/BF03196346

Chrastil, E. R. (2012). Neural evidence supports a novel framework for spatial navigation. *Psychonomic Bulletin & Review*. https://doi.org/10.3758/s13423-012-0351-610.3758/s13423-012-0351-6. Advance online publication.

Chrastil, E. R., & Warren, W. H. (2014). From cognitive maps to cognitive graphs. *PLOS ONE, 9*(11), e112544.

Clark, B. J., Simmons, C. M., Berkowitz, L. E., & Wilber, A. A. (2018). The retrosplenial–parietal network and reference frame coordination for spatial navigation. *Behavioral Neuroscience, 132*(5), 416–429. https://doi.org/10.1037/bne0000260

Credé, S., Thrash, T., Hölscher, C., & Fabrikant, S. I. (2019). The acquisition of survey knowledge for local and global landmark configurations under time pressure. *Spatial Cognition & Computation, 19*(3), 190–219. https://doi.org/10.1080/13875868.2019.1569016

Epstein, R. A., Patai, E. Z., Julian, J. B., & Spiers, H. J. (2017). The cognitive map in humans: Spatial navigation and beyond. *Nature Neuroscience, 20*, 1504–1513. https://doi.org/10.1038/nn.4656

Fitting, S., Wedell, D. H., & Allen, G. L. (2005). Memory for spatial location: Influences of environmental cues and task field rotation. In A. G. Cohn & D. M. Mark (Eds.), *Spatial information theory* ([Lecture Notes in Computer Science, vol. 3693], pp. 459–474). Springer. https://doi.org/10.1007/11556114_29

Fitting, S., Wedell, D. H., & Allen, G. L. (2007). Memory for spatial location: Cue effects as a function of field rotation. *Memory & Cognition, 35*(7), 1641–1658. https://doi.org/10.3758/bf03193498

Fitting, S., Wedell, D. H., & Allen, G. L. (2009). Cue effects on memory for location when navigating spatial displays. *Cognitive Science, 33*(7), 1267–1300. https://doi.org/10.1111/j.1551-6709.2009.01056.x

Gallistel, C. R. (1990). *The organization of learning*. MIT Press.

Gardony, A., Brunyé, T. T., Mahoney, C. R., & Taylor, H. A. (2011). Affective states influence spatial cue utilization during navigation. *Presence: Teleoperators and Virtual Environments, 20*(3), 223–240. https://doi.org/10.1162/pres_a_00046

Geiger, S., Gillner, S., & Mallot, H. A. (1997). Global versus local cues for route finding in virtual environments. *Perception, 26*(1_Suppl), 155–155. https://doi.org/10.1068/v970326

Hermer, L., & Spelke, E. S. (1994). A geometric process for spatial reorientation in young children. *Nature, 370*(6484), 57–59. https://doi.org/10.1038/370057a0

Huttenlocher, J., Hedges, L. V., & Duncan, S. (1991). Categories and particulars: Prototype effects in estimating spatial location. *Psychological Review, 98*(3), 352–376. https://doi.org/10.1037/0033-295x.98.3.352

Jacobs, L. F., & Menzel, R. (2014). Navigation outside of the box: What the lab can learn from the field and what the field can learn from the lab. *Movement Ecology, 2*(1), 3. https://doi.org/10.1186/2051-3933-2-3

Jacobs, W. J., Laurance, H. E., & Thomas, K. G. F. (1997). Place learning in virtual space I: Acquisition, overshadowing, and transfer. *Learning and Motivation, 28*(4), 521–541. https://doi.org/10.1006/lmot.1997.0977

Jansen-Osmann, P. (2002). Using desktop virtual environments to investigate the role of landmarks. *Computers in Human Behavior, 18*(4), 427–436. https://doi.org/10.1016/s0747-5632(01)00055-3

Jansen-Osmann, P., & Fuchs, P. (2006). Wayfinding behavior and spatial knowledge of adults and children in a virtual environment. *Experimental Psychology, 53*(3), 171–181. https://doi.org/10.1027/1618-3169.53.3.171

Jansen-Osmann, P., Schmid, J., & Heil, M. (2007). Wayfinding behavior and spatial knowledge of adults and children in a virtual environment: The role of the environmental structure. *Swiss Journal of Psychology, 66*(1), 41–50. https://doi.org/10.1024/1421-0185.66.1.41

Jellinger, K. A. (2000). Wayfinding behavior, cognitive mapping and other spatial processes. *European Journal of Neurology, 7*(5), 590–591. (Original work published 1999 by Reginald G. Golledge, Ed., The Johns Hopkins University Press).

Kelly, J. W., McNamara, T. P., Bodenheimer, B., Carr, T. H., & Rieser, J. J. (2008). The shape of human navigation: How environmental geometry is used in maintenance of spatial orientation. *Cognition, 109*(2), 281–286. https://doi.org/10.1016/j.cognition.2008.09.001

Klippel, A., & Winter, S. (2005). Structural salience of landmarks for route directions. In A. G. Cohn & D. M. Mark (Eds.), *Spatial information theory* ([Lecture Notes in Computer Science, vol. 3693], pp. 347–362). Springer.

Lin, C.-T., Huang, T.-Y., Lin, W.-J., Chang, S.-Y., Lin, Y.-H., Ko, L.-W., ..., Chang, E. C. (2012). Gender differences in wayfinding in virtual environments with global or local landmarks. *Journal of Environmental Psychology*, *32*(2), 89–96. https://doi.org/10.1016/j.jenvp.2011.12.004

Löwen, H., Krukar, J., & Schwering, A. (2019). Spatial learning with orientation maps: The influence of different environmental features on spatial knowledge acquisition. *ISPRS International Journal of Geo-Information, 8*(3), 149. https://doi.org/10.3390/ijgi8030149

Lynch, K. (1960). *The image of the city*. Technology Press.

Maguire, E. A., Burgess, N., Donnett, J. G., Frackowiak, R. S. J., Frith, C. D., & O'Keefe, J. (1998). Knowing where and getting there: A human navigation network. *Science, 280*(5365), 921–924. https://doi.org/10.1126/science.280.5365.921

Meilinger, T., Schulte-Pelkum, J., Frankenstein, J., Berger, D., & Bülthoff, H. H. (2015). Global landmarks do not necessarily improve spatial performance in addition to bodily self-movement cues when learning a large-scale virtual environment. *International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments, ICAT-EGVE 2015*, 25–28. https://doi.org/10.2312/egve.20151306

Merhav, M., & Fisher-Gewirtzman, D. (2023). How pathways' configuration impacts wayfinding in young and older adults. *Journal of Environmental Psychology, 90*, 1–7. https://doi.org/10.1016/j.jenvp.2023.102065

Montello, D. R., & Raubal, M. (2013). Functions and applications of spatial cognition. In D. Waller & L. Nadel (Eds.), *Handbook of spatial cognition* (pp. 249–264). American Psychological Association.

Newman, E. L., Caplan, J. B., Kirschen, M. P., Korolev, I. O., Sekuler, R., & Kahana, M. J. (2007). Learning your way around town: How virtual taxicab drivers learn to use both layout and landmark information. *Cognition, 104*(2), 231–253. https://doi.org/10.1016/j.cognition.2006.05.013

Peer, M., & Epstein, R. A. (2025). Cognitive maps for hierarchical spaces in the human brain. *Cerebral Cortex, 35*(9), bhaf261. https://doi.org/10.1093/cercor/bhaf261

Ruddle, R. A., Volkova, E., Mohler, B., & Bülthoff, H. H. (2010). The effect of landmark and body-based sensory information on route knowledge. *Memory & Cognition, 39*(4), 686–699. https://doi.org/10.3758/s13421-010-0054-z

Sandstrom, N. J., Kaufman, J., & Huettel, S. A. (1998). Males and females use different distal cues in a virtual environment

navigation task. *Cognitive Brain Research, 6*(4), 351–360. https://doi.org/10.1016/s0926-6410(98)00002-0

Sorrows, M. E., & Hirtle, S. C. (1999). The nature of landmarks for real and electronic spaces. In C. Freksa & D. M. Mark (Eds.), *Spatial Information Theory: Cognitive and Computational Foundations of Geographic Information Science* (pp. 37–50). Springer.

Steck, S. D., & Mallot, H. A. (2000). The role of global and local landmarks in virtual environment navigation. *Presence: Teleoperators and Virtual Environments, 9*(1), 69–83. https://doi.org/10.1162/105474600566628

Strickrodt, M., Bülthoff, H. H., & Meilinger, T. (2019). Memory for navigable space is flexible and not restricted to exclusive local or global memory units. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 45*(6), 993–1013. https://doi.org/10.1037/xlm0000624

Thorndyke, P. W., & Hayes-Roth, B. (1982). Differences in spatial knowledge acquired from maps and navigation. *Cognitive Psychology, 14*, 560–589. https://doi.org/10.1016/0010-0285(82)90019-6

Wang, R., & Spelke, E. (2002). Human spatial representation: Insights from animals. *Trends in Cognitive Sciences, 6*(9), 376–382. https://doi.org/10.1016/s1364-6613(02)01961-7

Weisberg, S. M., & Newcombe, N. S. (2018). Cognitive maps: Some people make them, some people struggle. *Current Directions in Psychological Science, 27*(4), 220–226. https://doi.org/10.1177/0963721417744521

Weniger, G., Ruhleder, M., Wolf, S., Lange, C., & Irle, E. (2009). Egocentric memory impaired and allocentric memory intact as assessed by virtual reality in subjects with unilateral parietal cortex lesions. *Neuropsychologia, 47*(1), 59–69. https://doi.org/10.1016/j.neuropsychologia.2008.08.018

Wiener, J. M., & Mallot, H. A. (2003). 'Fine-to-coarse' route planning and navigation in regionalized environments. *Spatial Cognition and Computation, 3*(4), 331–358.

Wolbers, T., & Wiener, J. M. (2014). Challenges for identifying the neural mechanisms that support spatial navigation: The impact of spatial scale. *Frontiers in Human Neuroscience*, *8*. https://doi.org/10.3389/fnhum.2014.00571

Wolbers, T., & Büchel, C. (2005). Dissociable retrosplenial and hippocampal contributions to successful formation of survey representations. *Journal of Neuroscience, 25*(13), 3333–3340. https://doi.org/10.1523/JNEUROSCI.4705-04.2005

Wolbers, T., & Hegarty, M. (2010). What determines our navigational abilities? *Trends in Cognitive Sciences, 14*(3), 138–146. https://doi.org/10.1016/j.tics.2010.01.001

Yesiltepe, D., Ozbil Torun, A., Coutrot, A., Hornberger, M., Spiers, H., & Conroy Dalton, R. (2020). Computer models of saliency alone fail to predict subjective visual attention to landmarks during observed navigation. *Spatial Cognition & Computation, 21*(1), 39–66. https://doi.org/10.1080/13875868.2020.1830993

Yesiltepe, D., Conroy Dalton, R., & Ozbil Torun, A. (2021). Landmarks in wayfinding: A review of the existing literature. *Cognitive Processing, 22*(3), 369–410. https://doi.org/10.1007/s10339-021-01012-x

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.